SUDHANSHU PATIL

AI ENGINEER

+91 72489 13660 | sudhanshupatil588@gmail.com | Nagpur, Maharashtra, India | Open to Pune, Hyderabad, Bangalore, Mumbai, Remote
LinkedIn | GitHub | Portfolio | LeetCode Knight (Top 5%) | Immediate Joiner

SUMMARY

AI Engineer with 1+ year owning production AI systems end to end as the sole AI engineer for banking and enterprise clients, including Bank of India: agentic RAG with human-in-the-loop review, MCP tool servers, guardrails and evaluation on Azure AI Foundry and AWS. Full-stack delivery (FastAPI, React, PostgreSQL, Redis) and shipped products with real users.

WORK EXPERIENCE

AI Engineer (Freelance) **Jul 2026 – Present**
Self-employed, Remote

- For a startup client (Sep – Oct 2026), built multi-tenant MCP servers for Claude, ChatGPT and Gemini with OAuth 2.1 + PKCE, dynamic client registration and per-tenant resource discovery (RFC 9728).
- Added approval checks on every tool call and per-call metered credits billed in one race-safe UPDATE, so balances never go negative.
- Deployed on AWS ECS Fargate (ALB, SSM, zero-downtime deploys); shipped 19 PRs in a week on a WhatsApp CRM (Meta Cloud API).
- Independently built and shipped agentic RAG and AI tools with hybrid retrieval, evaluation and guardrails (Waypoint, MediClaim, Glance).

AI Solutions Developer **Aug 2025 – Jun 2026**
SimpleWorks, Nagpur, India

- Built a hub-and-spoke agentic RAG framework (orchestrator, intent recogniser, dynamic query splitting, math sandbox) that passed QA for BFSI clients including Bank of India, with up to 2 levels of human-in-the-loop review.
- Architected a cloud-hosted FastMCP server exposing 10+ production tools across 9 CRM modules to Claude, Gemini CLI and internal apps (40k-token memory), and designed the MVP for an internal AI platform with multi-DB connectors, RBAC, Langfuse tracing and @mention queries.
- Cut PII-redaction latency 94% (8s to 0.5s) by replacing a spaCy transformer with a fine-tuned DistilBERT NER model, with triple-verification guardrails that keep excluded fields out of every LLM payload.
- Solo-led Python 3.11 to 3.13 and LlamaIndex migrations on Azure AI Foundry with zero rollbacks on QA and PSB production, including offline wheel packaging for air-gapped banking systems, unblocking GPT-5-series adoption.
- Owned architecture through production deployment as the sole AI engineer: 8,000+ lines of production code, 5+ major SQL fixes and 20+ QA defects resolved.

Associate Data Scientist (Intern) **Jan 2025 – Jul 2025**
Novartis, Hyderabad, India

- Awarded a full-time Pre-Placement Offer (PPO) for performance; declined to join SimpleWorks.
- Built an on-prem LLM chatbot on internal R-coding docs with OSS-comparable answers and zero clinical data leaving the network.
- Automated work for 60+ Data Managers with VBA and Office Scripts (auto-filing, input guardrails); replaced SAS reporting with RShiny.

PROJECTS

Waypoint: AI Travel Concierge Demo | Live

- Built a Google ADK agent on Groq that turns a plain-English trip request into three itineraries from real Places, Routes and fare data only; the LLM extracts intent while dates, deadlines and budgets are computed in code, and plans that fail verification are never stored.
- Production-hardened on free tiers: API spend metered to stop at 80% of each allowance, idempotent leased workers for Telegram flight and weather alerts, a fix for Groq rejecting every conversation at turn 2, and ~1,100 layered tests with mutation checks.

MediClaim: Agentic RAG for Insurance Claims Code | Demo

- Designed a LangGraph router that picks vector, GraphRAG or hybrid retrieval per query, scoping costly graph extraction to relationship-dense policies so the system scales toward 100k documents; version-aware ingestion means superseded policies are never cited.
- Made citations structural: schema-constrained generation emits the chunk IDs it used, powering grounding checks, an LLM judge and human review of high-risk claims, on a QLoRA-tuned 3B Llama with a RAGAS regression suite and Langfuse tracing.

Glance: Open-Source AI Assistant for Portfolios Code | Live

- Zero-dependency widget that reads the host site in the visitor's browser, including client-rendered React pages and project-labelled links, so site owners never write a knowledge base.
- Fits Groq's free 8K-tokens-per-minute limit with a shared context budget and model fallback; origin-locked API with per-visitor rate limits; one-script-tag setup that powers my portfolio.

SKILLS

AI / ML: LLMs | RAG | Agentic AI | LangGraph | LlamaIndex | LangChain | MCP | NER | QLoRA/PEFT | PyTorch | RAGAS | Guardrails
Languages and Backend: Python | C++ | SQL | FastAPI | REST APIs | Async (ARQ) | Pydantic | OAuth 2.1 | React
Data and Databases: PostgreSQL | MySQL | Redis | SQLite | Qdrant | ChromaDB | Neo4j
DevOps and Cloud: Docker | Kubernetes | GitHub Actions | Git | Linux | Nginx | AWS (Bedrock, ECS Fargate) | GCP | Azure AI Foundry | Langfuse

EDUCATION

BITS Pilani, Hyderabad Campus **Oct 2021 – Jul 2025**
Bachelor of Engineering in Electronics and Communication Engineering
Minor in Computing and Intelligence

HONORS AND AWARDS

INSPIRE-SHE Scholar, Top 1% in Maharashtra HSC (97.5%) | Naukri EROH University, 2nd Runner-up | LeetCode Knight, Top 5%

CERTIFICATIONS

2026: Kubernetes LFS158 (Linux Foundation) | Docker Foundations Professional | Databricks Fundamentals | Oracle Agentic AI Certified Professional | Oracle Agentic AI Foundations Associate
2024: Oracle Gen AI Certified Professional | **Courses:** Multi AI Agent Systems with CrewAI | Mastering AutoGen